Report 1: The Delay–Doppler Grid and Its Timing

EE451 — Zak Transform / Delay–Doppler Communication
Simulation S1

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1 What this report is about

- Before any transform, we must know *what the grid is* and *how it sits in time*.
- This report answers: how many symbols are in a frame, how far apart are the grid points in delay and Doppler, and at which sample of the DAC does each symbol's pulse sit.
- The simulation draws these relations so the formulas become something you can see.

2 The concept in brief

- **Two ways a wireless echo changes a signal.** *Delay*: how late it arrives. *Doppler*: how fast its phase rotates because of motion. The delay–Doppler (DD) domain uses exactly these two quantities as axes.
- **The DD grid** $X[m, n]$: a table of M rows (delay bins, index m) and N columns (Doppler bins, index n). Each cell holds one complex symbol, so a frame carries MN symbols.
- **Delay spacing.** $\Delta\tau = T_s = 1/B$. More bandwidth $\Rightarrow$ shorter pulses $\Rightarrow$ we can separate echoes that are closer together.
- **Zak period.** One block of M delay bins (one column of the grid) takes $T = M\,T_s$. A frame is N such periods: $T_{\text{frame}} = N\,T$.
- **Doppler spacing.** $\Delta\nu = 1/T_{\text{frame}}$. Watching for longer $\Rightarrow$ we can tell finer speed differences. The total Doppler range is $1/T$.
- **Hardware samples.** The DAC takes D samples per delay bin: $T_{\text{samp}} = T_s/D$ (D=1: critical sampling, D>1: oversampled). One Zak period therefore holds $P = MD$ samples.
- **Where a symbol lives in time.** The pulse for the symbol at delay bin m in period q is centred at sample

$$a = D\,m + P\,q.$$

  m moves you a little (steps of D), q moves you a whole period (steps of P).

One-line memory aid: bandwidth fixes the delay step $1/B$; frame length fixes the Doppler step $1/T_{\text{frame}}$; $a = Dm + Pq$ tells you where each symbol sits on the time axis.

Quantity	Formula	Meaning
Delay step	$T_s = 1/B$	width of one delay bin
Zak period	$T = MT_s$	time of one block of M bins
Frame length	$T_{\text{frame}} = NT$	N periods
Doppler step	$\Delta\nu = 1/T_{\text{frame}}$	width of one Doppler bin
Doppler range	$1/T$	N bins $\times$ $\Delta\nu$
DAC sample time	$T_{\text{samp}} = T_s/D$	D samples per delay bin
Samples per period	$P = MD$	
Pulse centre	$a = Dm + Pq$	symbol (m, q)

3 What was simulated

- A small random DD grid (M=16, N=8) to see what a sparse frame looks like.
- The pulse-centre positions on the sample axis for a small frame (M=8, N=4, D=4).
- A map of the sample index $a = Dm + Pq$ for every grid cell (M=8, N=4, D=2).
- Delay and Doppler steps against bandwidth and against N, and a numerical table for the example M=N=128, B=500 MHz.

4 Results and discussion

4.1 The DD grid

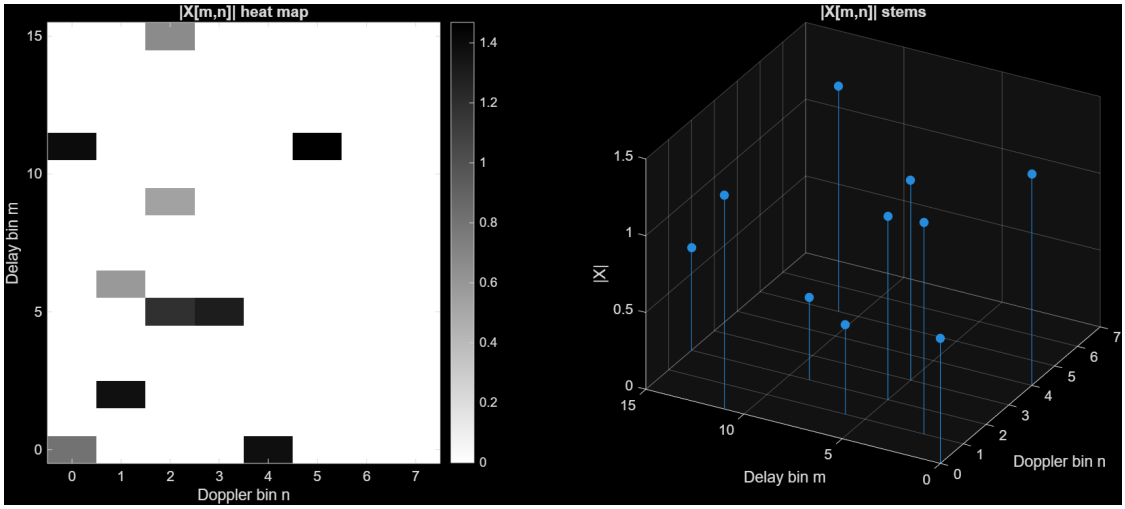


Figure 1: A random sparse DD frame (M=16, N=8): heat map (left) and stem view (right).

- Each coloured cell is one symbol; here 10 of the 128 cells are occupied.
- **Expected:** the position of a symbol is its (delay, Doppler) address, and its height is its amplitude. **Observed:** the heat map and the stem view both show the same 10 symbols.
- Takeaway: the DD grid is just a memory table; a transmitter fills it with symbols, and the inverse Zak transform (IZak) later turns it into a waveform.

4.2 Where the pulses sit in time

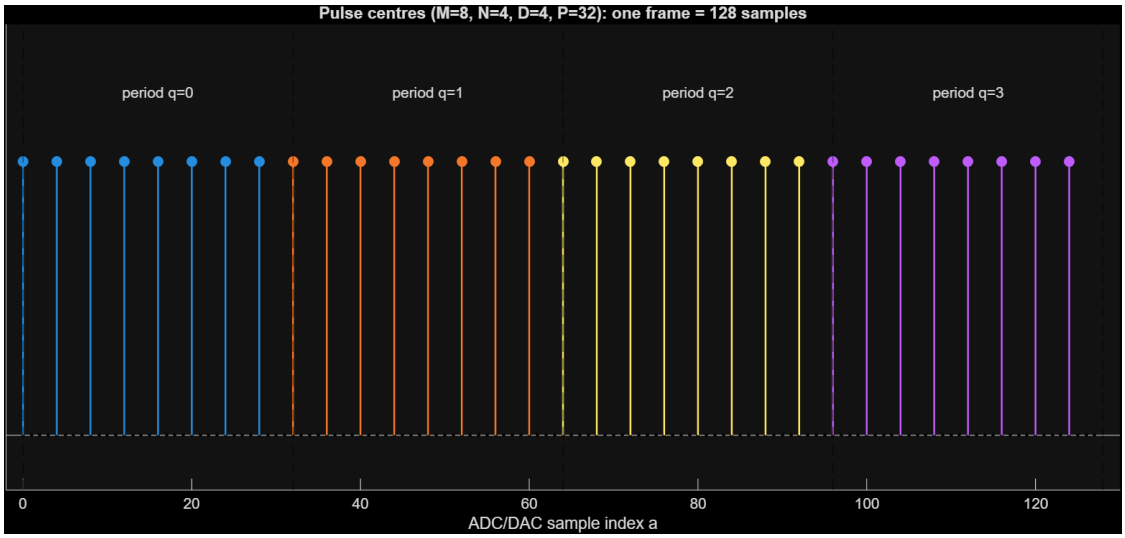


Figure 2: Pulse centres for M=8, N=4, D=4 (P=32): one frame is 128 samples.

- **Expected:** M=8 centres per period, spaced D=4 samples apart, in N=4 periods of P=32 samples each, so $MND = 128$ samples per frame.
- **Observed:** exactly that: 8 evenly spaced dots in each of 4 colour-coded periods, the next period starting at samples 32, 64 and 96.
- Takeaway: a Zak period is simply a block of P samples; a frame is N blocks in a row.

4.3 The index rule $a = Dm + Pq$

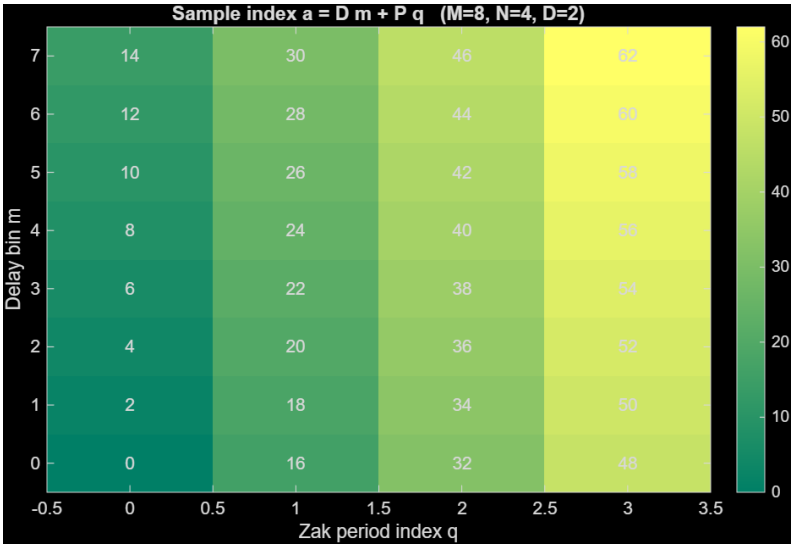


Figure 3: Sample index of the pulse centre for each cell (M=8, N=4, D=2, P=16).

- **Expected:** going up one delay bin adds D=2; going right one period adds P=16.
- **Observed:** the first column reads $0, 2, 4, \ldots, 14$; the next starts at 16; and e.g. (m=3, q=2) $\mapsto$ $6 + 32 = 38$, which is the value shown in the map.
- Takeaway: each column of the grid is one block of time; rows are positions inside the block.

4.4 Resolution against bandwidth and frame size

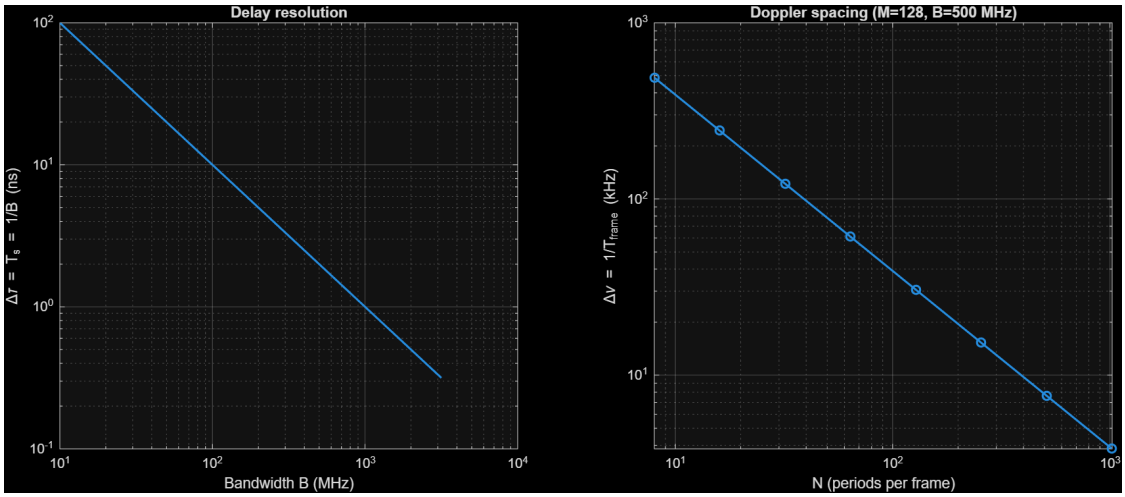


Figure 4: Left: delay step $T_s = 1/B$ against bandwidth. Right: Doppler step $1/T_{\text{frame}}$ against N (M=128, B=500 MHz).

- **Expected:** both are inverse laws, i.e. straight lines of slope -1 on log–log axes.
- **Observed:** both curves are straight lines of slope -1. For example, B=500 MHz gives $T_s = 2$ ns, and N=128 gives $\Delta\nu \approx 30.5$ kHz.
- Takeaway: to see finer delays use more bandwidth; to see finer Doppler use a longer frame (larger N).

Table 1: Numbers for $M = N = 128$, $B = 500$ MHz.

Quantity	Value
Symbols per frame MN	16 384
Delay step T_s	2 ns
Zak period T	256 ns
Frame length T_{frame}	32.768 μs
Frame rate	30 517.6 frames/s
Doppler step $\Delta\nu$	30.52 kHz
Doppler range $1/T$	3.906 MHz
DAC side	
$D = 1$: T_{samp}, P, samples per frame	2 ns, 128, 16 384
$D = 4$: T_{samp}, P, samples per frame	0.5 ns, 512, 65 536
Sample rate for $D = 4$	2 GS/s

5 Summary

- The frame holds MN symbols; delay step $= 1/B$, Doppler step $= 1/(NMT_s)$.
- In time, a frame is N blocks of $P = MD$ samples, and symbol (m, q) is centred at $a = Dm + Pq$.
- All printed values and plots agree with the formulas; nothing in S1 deviates from expectation.

Appendix: details and assumptions

- $\Delta\tau = 1/B$ is stated in the course slides (p. 9). The Doppler step $1/T_{\text{frame}}$ and the range $1/T$ are *derived* from the N-point DFT taken over the N Zak periods; they are not written explicitly in the slides.
- The frame is treated as periodic, so sample indices wrap around modulo MND.
- Figures were produced in MATLAB (`S1_grid_timing.m`, parameters in `otfs_params.m`); the figures use MATLAB's dark theme.